

# Ramesh Sirimalla

Forward Deployed Engineer — Real-Time Data Streaming & Generative AI

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## SUMMARY

Forward Deployed Engineer with 15+ years of experience building production-grade real-time data platforms, APIs, and Generative AI systems across healthcare, financial services, and public-sector programs. Experienced in architecting multi-agent workflows, enterprise RAG systems, and large-scale Kafka streaming platforms, and in working directly with business and third-party partners to take solutions into production.

## CORE SKILLS

- Streaming & Messaging:** Confluent Kafka, Kafka Streams, Kafka Connect, KSQL, Apache Flink, Redpanda, Azure Service Bus, Spark Streaming, StreamSets
- Generative & Agentic AI:** LLMs, Prompt Engineering, RAG, LangChain, LangGraph, OpenAI API, Amazon Bedrock, Model Context Protocol, LLM Evaluation, Guardrails, FAISS, Pinecone, Chroma
- Programming & Frameworks:** Java, Python, SQL, C++, Spring Boot, Spring Integration, Spring Batch, Hibernate, PySpark, Pandas
- Data, Cloud & DevOps:** Oracle, SQL Server, MySQL, MongoDB, Redis, Google Spanner, Hadoop, AWS, Docker, Kubernetes (AKS, GKE), Jenkins, GitHub Actions, Splunk, Grafana

## EXPERIENCE

- Forward Deployed Engineer**  
CVS Health — Dallas, TX

Dec 2023 — Present

  - Led design and development of real-time streaming applications delivering patient and customer notifications across SMS, email, push, and IVR.
  - Built Kafka Streams applications with Spring Boot to prevent duplicate messages and ensure delivery within the legal time window.
  - Built a POC decoupling RESTful services with Azure Service Bus for prescription payment audits, with computation in Apache Flink; delivered further POCs on Redpanda, Kafka, and Flink.
  - Lead engineering teams building healthcare solutions and mentor junior engineers.
- Senior Lead Engineer**  
CVS Health — Dallas, TX

Jun 2020 — Dec 2023

  - Built Spring Boot services for a real-time compliance-check API and a Patient Preferences API for CVS.com users.

- Built Spring Integration applications on Confluent Kafka to capture events in real time and update downstream CVS systems.
- Built nightly schedulers loading ROCM configuration from Oracle to MongoDB, and set up C3 and Splunk alerting for this Tier 1 orchestration hub.

## Senior Data Engineer

Jan 2019 — May 2020

### Early Warning Services (Zelle) — Scottsdale, AZ

- Developed Spring Boot services with Kafka Connect and KSQL storing audit transactions for real-time dashboards.
- Designed complex KStream topologies to dedupe messages and join multiple topics into a single downstream payload.
- Built StreamSets pipelines for Mastercard reconciliation, and Grafana and Splunk dashboards for Visa and Zelle reporting.

## Kafka Lead Developer

Jun 2018 — Dec 2018

### CBRE — Dallas, TX

- Developed Kafka Connect, KSQL, and KStream services for real-time dashboards, plus Splunk and Grafana reporting.

## Big Data Developer

Apr 2015 — May 2018

### New York State (ITS) — Albany, NY

- Integrated Kafka and Spark Streaming for real-time analytics and built big data solutions on the Hadoop ecosystem.

## Senior Software Engineer

Apr 2013 — Mar 2015

### New York State Police (DCJS) — Albany, NY

## Senior Software Engineer

Nov 2011 — Mar 2013

### Pearson VUE — Iowa City, IA

## EDUCATION

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**Bachelor of Engineering, Computer Science** — MVSR Engineering College (Osmania University), Hyderabad, India, 2001 – 2005 (First Class)

## CERTIFICATIONS

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Confluent Fundamentals Accreditation — Confluent